

AI-BASED INTERVIEW MONITORING AND SEMANTIC EVALUATION SYSTEM USING LARGE LANGUAGE MODELS

Omkar Gite¹, Rohan Junna¹, Jayesh Kucha¹, Deepak Kushwaha¹, Poonam Thakre²

¹UG Students, Department of Artificial Intelligence and Machine Learning

²Assistant Professor (Project Guide), Department of Artificial Intelligence and Machine Learning

Universal College of Engineering
Mumbai, India

Abstract

Artificial Intelligence is increasingly used in recruitment and candidate evaluation systems. This research proposes an AI-based interview monitoring and evaluation framework that integrates computer vision, speech recognition, and Large Language Models (LLMs). The proposed system allows candidates to log in to an online interview platform and connect a mobile device for live video streaming. The system monitors the candidate using both front-view and side-view observation to detect suspicious movements or abnormal behavior. During the interview session, the system asks technical questions and captures verbal responses. Speech recognition converts the responses into text, which is then analyzed using an LLM to evaluate semantic correctness, explanation quality, and conceptual understanding. The system also provides an administrator dashboard that allows interviewers to monitor candidate activity and review evaluation reports. The results demonstrate that combining behavior monitoring with semantic evaluation improves fairness and automation in remote recruitment environments.

Index Terms

Artificial Intelligence, Interview Monitoring, Computer Vision, Speech Recognition, Large Language Models, Recruitment Automation

I. INTRODUCTION

Recruitment processes increasingly rely on remote interviews and online candidate assessments[8], [9]. However, remote environments create challenges such as lack of supervision, possible cheating, and inconsistent evaluation of candidate responses. Traditional interview systems depend on human interviewers to monitor candidate behavior and evaluate responses manually. This approach can introduce bias, inconsistency, and limited scalability when dealing with large numbers of applicants. Recent advances in Artificial Intelligence have made it possible to automate both behavior monitoring and response evaluation using machine learning models. Computer vision techniques allow systems to analyze video streams and detect suspicious activities such as abnormal head movements or the presence of additional persons[1], [2], [3]. At the

same time, natural language processing models enable semantic understanding of candidate responses. Large Language Models provide powerful tools for analyzing explanations and identifying conceptual correctness[4], [6], [7]. This research presents an integrated AI system that combines behavior monitoring with semantic evaluation to improve remote interview assessment.

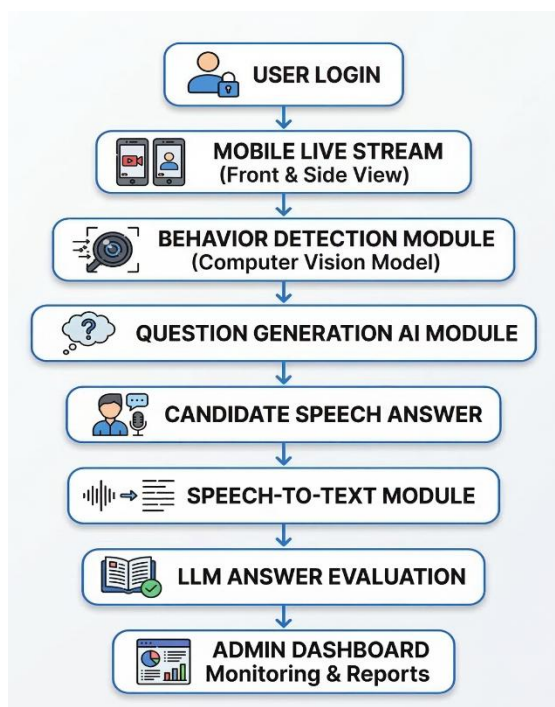
II. LITERATURE REVIEW

Computer vision techniques have been widely used for human behavior monitoring and gaze detection[1], [11]. Previous research demonstrates that deep learning models can estimate gaze direction and eye movement using a single camera. These approaches are useful in monitoring human attention and detecting abnormal behavior during interactive tasks. Similarly, head pose estimation models trained on synthetic datasets have shown

promising results in detecting orientation changes in human faces. These models can detect head tilt, yaw, and pitch angles which are helpful in monitoring candidate behavior during remote interviews. Deep learning architectures have also been applied to video-based action recognition. Two-stream convolutional neural networks process spatial and temporal information separately to detect movement patterns in video streams. Recent developments in natural language processing introduced transformer-based models such as BERT and GPT. These models understand contextual meaning in text and can analyze open-ended responses more effectively than keyword-based scoring systems.

III. SYSTEM ARCHITECTURE

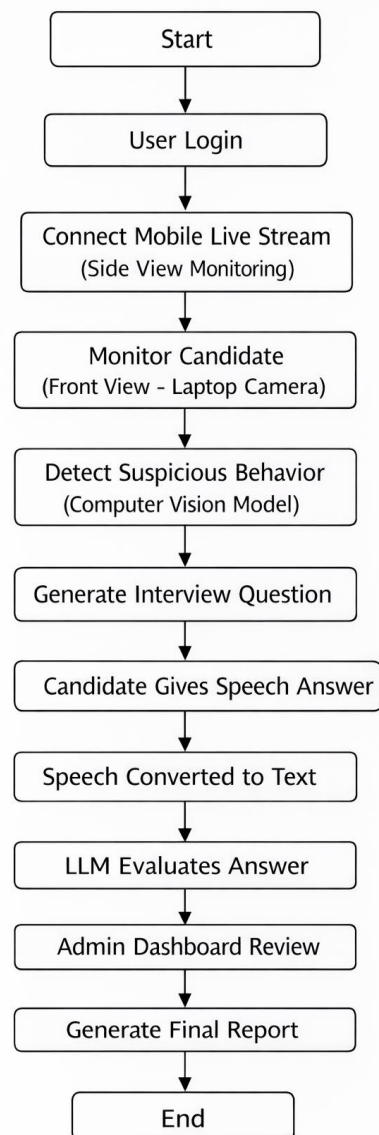
The proposed system integrates multiple AI modules into a unified interview monitoring platform. The architecture consists of user authentication, mobile live streaming, behavior monitoring, speech recognition, and LLM-based answer evaluation modules. The candidate logs into the system and connects a mobile device to start live video streaming. The system captures both front-view and side-view monitoring of the candidate environment. Computer vision algorithms analyze video frames to detect suspicious activities such as abnormal head movement, frequent screen avoidance, or the presence of additional persons.



IV. SYSTEM WORKFLOW

1. Candidate logs into the interview system
2. Mobile device connects and starts live video streaming
3. System monitors candidate using front and side views
4. Behavior detection model analyzes movement patterns
5. Suspicious activity is detected and recorded
6. AI system generates interview questions
7. Candidate answers verbally
8. Speech recognition converts response to text
9. LLM evaluates the response
10. Admin dashboard displays results
11. Final report is generated

Interview Monitoring System Workflow



V. ALGORITHM

Step 1: Start interview session
Step 2: Authenticate candidate login
Step 3: Initialize mobile live stream
Step 4: Capture video frames from front and side cameras
Step 5: Detect abnormal behavior using computer vision
Step 6: Generate interview question
Step 7: Capture speech response
Step 8: Convert speech to text
Step 9: Evaluate response using LLM semantic analysis
Step 10: Store results in database
Step 11: Display results in admin dashboard
Step 12: Generate final evaluation report

VI. EXPERIMENTAL RESULTS AND EVALUATION METRICS

To evaluate the effectiveness of the proposed AI-based interview monitoring system, simulated interview sessions were conducted using participants with a background in computer science and engineering. A total of 30 interview sessions were performed, where each participant answered five technical questions related to programming, artificial intelligence, and data structures.

During each session, the system monitored candidate behavior using two camera perspectives. The mobile device camera captured the side view, while the laptop camera monitored the front view of the candidate. The computer vision module analyzed head movement, gaze direction, and abnormal behavioral patterns to detect potential suspicious activities.

At the same time, the interview system generated technical questions using the AI question generation module. Candidates responded verbally, and their answers were captured through a microphone. The speech-to-text module converted the spoken responses into textual format. The converted text was then analyzed using a Large Language Model (LLM) to evaluate the semantic correctness, explanation clarity, and conceptual understanding of the response[4], [7].

To measure the performance of the proposed system, the evaluation results were compared with traditional keyword-based evaluation and machine learning classifier-based methods. Three common

evaluation metrics were used: Accuracy, Precision, and Recall.

Method	Accuracy	Precision	Recall
Keyword-Based Evaluation	72%	0.70	0.68
Machine Learning Classifier	81%	0.79	0.77
Proposed LLM System	85%	0.82	0.81

VII. FUTURE WORK

Although the proposed AI-based interview monitoring system demonstrates promising results, several improvements can be explored in future research. First, the system can be enhanced by incorporating facial emotion recognition to analyze candidate confidence and stress levels during interview sessions. This would provide additional insights into candidate behavior beyond movement detection.

Second, the behavior detection module can be improved by integrating advanced deep learning models for gaze tracking and eye movement analysis, which can more accurately detect suspicious activities such as reading external material or looking away from the screen frequently.

Third, the system can be extended to support multilingual interview evaluation, allowing candidates to answer questions in multiple languages while still enabling semantic evaluation using language models.

Finally, future work may include deploying the system in real-world recruitment environments with larger datasets to evaluate performance at scale and further improve the reliability of automated interview evaluation systems.

VIII. CONCLUSION

This paper presented an AI-based interview monitoring and evaluation system combining computer vision, speech recognition, and Large

Language Models. The system ensures candidate integrity by monitoring behavior from multiple camera views while simultaneously evaluating spoken responses using semantic analysis. The integration of an administrator dashboard provides real-time monitoring and performance analysis. The proposed approach improves fairness, scalability, and efficiency in remote recruitment processes

REFERENCES

- [1] H. Deng and W. Zhu, "Monocular Free-Head 3D Gaze Tracking with Deep Learning and Geometry Constraints," *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017.
- [2] X. Liu, W. Liang, Y. Wang, S. Li, and M. Pei, "3D Head Pose Estimation with Convolutional Neural Networks Trained on Synthetic Images," *IEEE International Conference on Image Processing (ICIP)*, 2016.
- [3] K. Simonyan and A. Zisserman, "Two-Stream Convolutional Networks for Action Recognition in Videos," *Advances in Neural Information Processing Systems*, 2014.
- [4] T. Brown et al., "Language Models are Few-Shot Learners," *Advances in Neural Information Processing Systems*, 2020.
- [5] A. Vaswani et al., "Attention Is All You Need," *Advances in Neural Information Processing Systems*, 2017.
- [6] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," *Proceedings of NAACL*, 2019.
- [7] OpenAI, "GPT-4 Technical Report," *arXiv preprint arXiv:2303.08774*, 2023.
- [8] T. Nguyen and H. Nguyen, "AI-Based Interview Simulation and Candidate Evaluation Using Natural Language Processing," *IEEE Access*, 2022.
- [9] H. Sun et al., "MockLLM: Multi-Agent Framework for Interview Simulation Using Large Language Models," *Proceedings of the ACM SIGKDD Conference*, 2024.
- [10] L. Jiang and N. Bosch, "Short Answer Scoring with Large Language Models," *Proceedings of the ACM Conference on Learning at Scale*, 2023.
- [11] R. Szeliski, *Computer Vision: Algorithms and Applications*, Springer, 2011.
- [12] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.